

TrES-1b

R-1835

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_150719 · created 2026-08-03T06:19:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457222.8584 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.07 +/- 0.053
Transit depth (Rp/Rs)^2	0.49 +/- 0.75 [%]
Orbital Inclination (inc)	89.77 +/- 1.41
Airmass coefficient 1 (a1)	0.76 +/- 0.45
Airmass coefficient 2 (a2)	0.02 +/- 0.47
Scatter in the residuals of the lightcurve fit is	49.23 %
Best Comparison Star	#2 - [105, 456]
Optimal Method	PSF photometry
Transit Duration (day)	0.0959 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.07 ± 0.053 vs 0.1358 (deviation 1.24σ)
Duration fit vs published	0.0959 d vs 0.1042 d (deviation 1.36σ)
Mid-time O–C	-2.9 min
Depth SNR	1.3
Residual scatter	49.23 %

Light curve

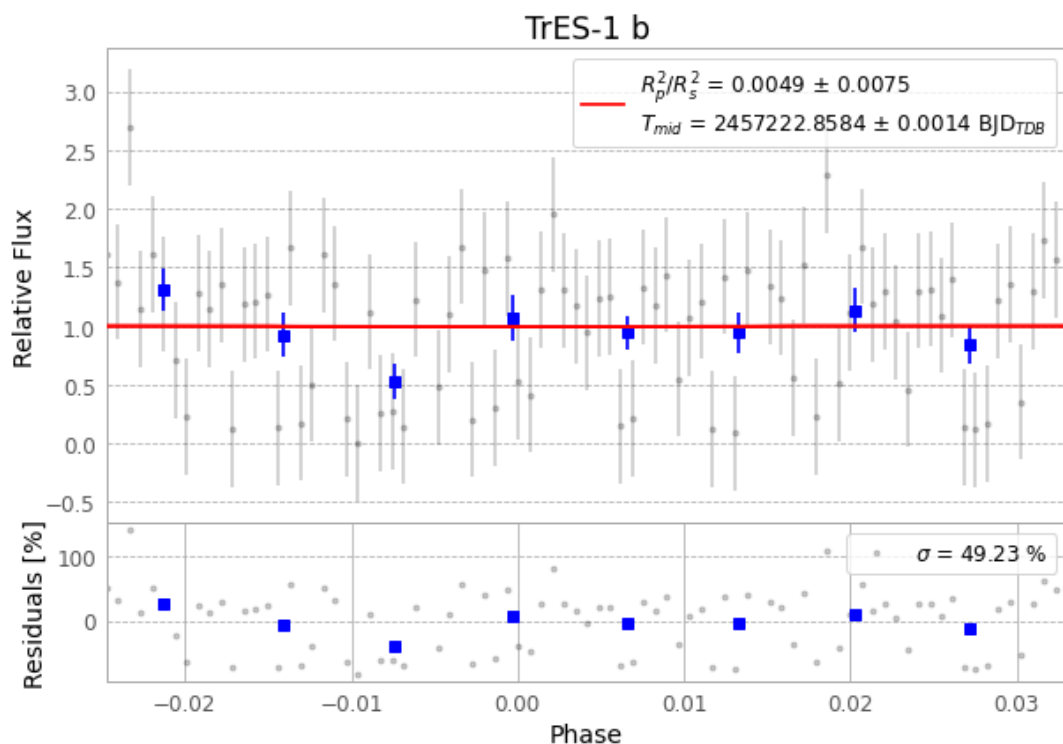


Figure 1 — FinalLightCurve_TrES-1 b_2015-07-18.png (SHA-256 99fa1e8daf0cf5db...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [125, 335], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cc0030fdfebef5ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), TRES-1150719064215.FITS → TRES-1150719105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-150719.log (78be93356281...), FinalLightCurve_TrES-1 b_2015-07-18.png (99fa1e8daf0c...), FinalLightCurve_TrES-1 b_2015-07-18.csv (c0e067cf00de...)

Compute resources

Wall time 230.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:19Z.